Ansh Roy

Robotics & Control Engineer

RESUME_1589

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Total Experience: 10 years

PROFESSIONAL SUMMARY

Robotics & Control Engineer with 10 years of experience in robotics, using ROS 2, Python, C++, MATLAB. Experienced in autonomous navigation, control loops and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Simulink, PID Control, C++, Motion Planning, LiDAR, SLAM, Python, Gazebo, ROS 2, PLC, MATLAB, Embedded Systems

Methods: Autonomous navigation, Control loops, Sensor fusion

PROFESSIONAL EXPERIENCE

Lead Robotics & Control Engineer | KoraLink Innovations | Jan 2023 - Present

- Validated 695 robot simulation scenarios with recorded sensor logs
- Lowered navigation recovery events by 12% using improved obstacle handling
- Reduced robotic path completion time by 18% through route and controller tuning
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Robotics & Control Engineer | HelioNex Labs | Jan 2019 - Dec 2022

- Reduced robotic path completion time by 22% through route and controller tuning
- Improved pick-position repeatability by 16% after calibration updates

Associate Robotics & Control Engineer | Urbanaut Engineering | Jul 2016 - Dec 2018

- Validated 328 robot simulation scenarios with recorded sensor logs
- Lowered navigation recovery events by 41% using improved obstacle handling

PROJECTS

Autonomous Mobile Robot | PID Control, ROS 2, C++, LiDAR

- Built ROS 2 nodes for localization, obstacle avoidance and route execution.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

Warehouse Mapping Trial | PLC, ROS 2, SLAM, Gazebo

- Benchmarked LiDAR SLAM configurations under moving obstacles.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

EDUCATION

B.Tech in Mechatronics Engineering, Western Plains Institute of Technology, 2016

Relevant Methods: Autonomous navigation, Control loops, Sensor fusion

CERTIFICATIONS

Control Systems Essentials (2018)